

AI GreetCraft

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AI GreetCraft: AI-Powered Greeting Card Figma Plugin

AI GreetCraft is an AI-powered Figma plugin designed to generate personalized greeting cards. It combines layout generation, user-uploaded photos, a custom AI-generated poem, and a thematic AI-generated background. The plugin enables users to create aesthetically pleasing and emotionally resonant cards directly within Figma, using a seamless and intuitive interface powered by GenAI technologies.

Integrated Card Creation Pipeline

Fully Integrated Approach

AlGreetCraft - fully integrated card creation pipeline, from personalized poem generation using GenAI to automated poem, background image, and layout generation.

Seamless User Experience

All functionalities are embedded directly within the Figma design environment, allowing for a more coherent and customizable user experience.

Related work

- **AI Card Craft** – Simple web-based card generation; lacks editing, no image upload.
- **Gemma 2 for Bengali Poetry** (*Acharya, 2024*) – Inspired our poem fine-tuning approach.
- **Hugging Face Diffusers** – Basis for text-to-image fine-tuning and background generation.
- **Figma Plugin API** – Enabled seamless integration into the design workflow.

System Components



Poem Generator

A fine-tuned LLM for occasion-specific poetry.



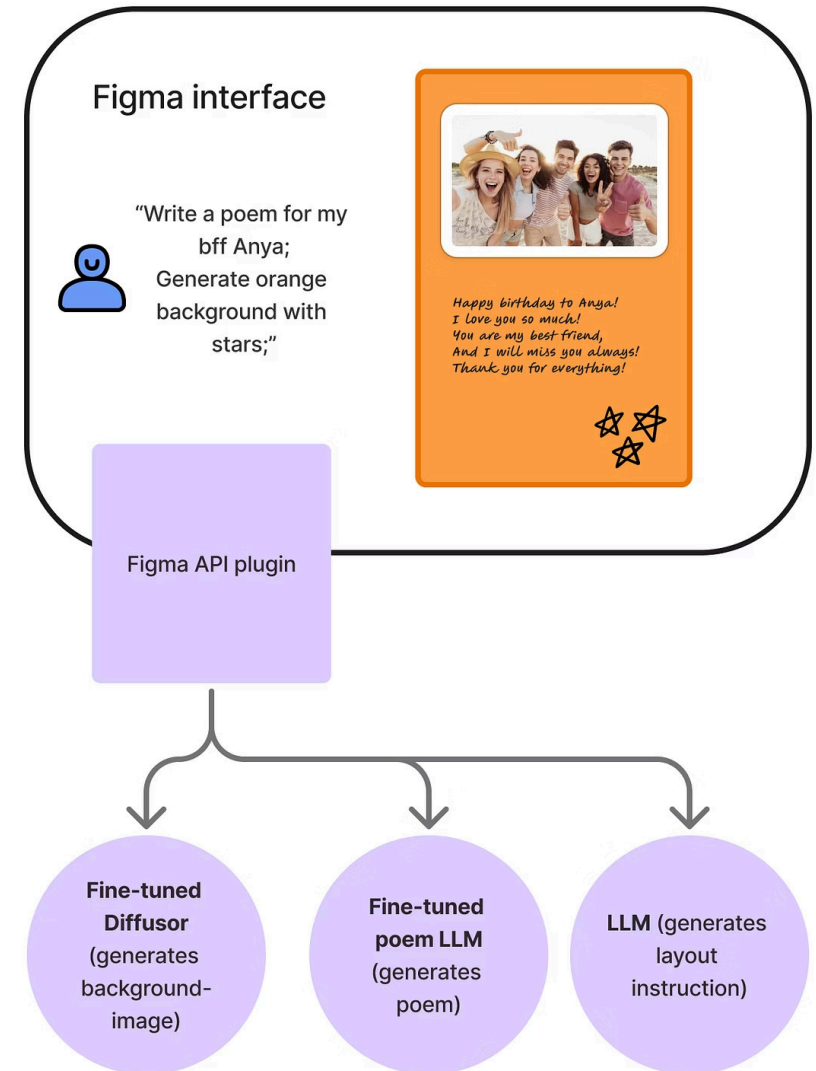
Background Generator

A custom-trained diffusion model for thematic background generation.



Figma Layout Composer

A plugin utilizing the Figma API to arrange visual elements dynamically.



Datasets

The system utilizes two datasets:

- For **Poems Generator** we parsed <https://www.wishafriend.com/poems/> and got 2,034 greeting poems annotated with themes and titles.
- For **Background Generator** we used a subset of 2000 images from the 5,640 images of Describable Textures Dataset (<https://www.robots.ox.ac.uk/~vgg/data/dtd/>).

a background with a chequered texture



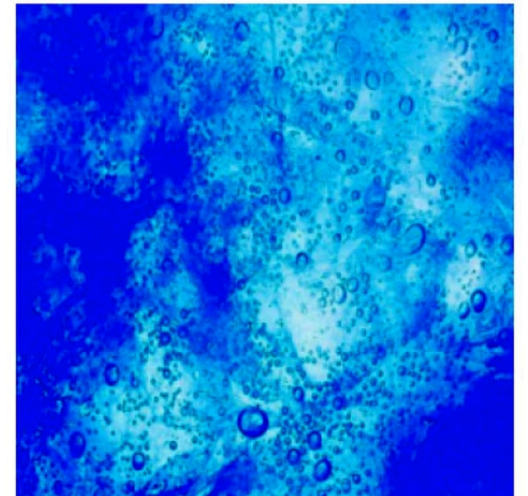
a background with a bumpy texture



a background with a braided texture



a background with a bubbly texture



Poem Generation Methodology

- 1

Fine-Tuning

using parameter-efficient finetuning (PEFT) and Low-Rank Adaptation (LoRA)
- 2

Training Dataset

of 2,034 curated samples, each containing a Title, Theme, and Poem.
- 3

Unsloth Ecosystem

used for experiments. It enables efficient training on consumer-grade hardware.

Multiple models were tested, after evaluation it was decided to use Gemma 2B:

Model	Parameters, size	# Epochs	Training Time	Loss	GPU RAM Usage	Qualitative Output Summary
T5-small	60M, 0.24GB	50	10 min	0.018	12.7GB	Overfits quickly; low coherence; fails to preserve prompt meaning.
GPT2-medium	345M, 1.42GB	3	80 min	0.52	12.7GB	Too generic; repetitive love-related texts; prompt ignored.
LLaMA3-1.8B + Unsloth	1.8B, 3.8GB	1	6 min	1.67	5.8GB	Moderate coherence; maintains partial prompt context; slightly verbose.
Gemma 2B + Unsloth	2.0B, 4.1GB	1	4 min	1.45	4.1GB	Most robust; prompt-aware; emotionally relevant; suitable for deployment.

```
instruction = "Generate a Father's Day poem for my dad"
themes = "strength, guidance, pride"
```

```
Generated Poem:
Dad you are strong

You have been guiding me since I was young

It's all because of your love in life

That everything gets sorted in time

I just want to thank you so much

For being there when I needed you the most

Happy Fathers day to you!
```

```
instruction = "Write a sweet poem to my best
friend to cheer her up"
themes = "support, happiness, shared memories"
```

```
Generated Poem:
The support you gave me when I needed it most

You were there whenever I was lost

I will never forget all our crazy times together

Our funny fights and our silly chatter

Your friendship has been so precious in life

There are many things that make us smile

But today something really dark seems to be there

So please don't worry girl, everything will be fine

Everything will turn out well!
```


Background Image Generation

1

Fine tuning of Stable Diffusion (CompVis/stable-diffusion-v1-4) for text-to-image generation

2

Trained on 2000/5640 images of DTD dataset

3

Used accelerate, Hugging Face Diffusers

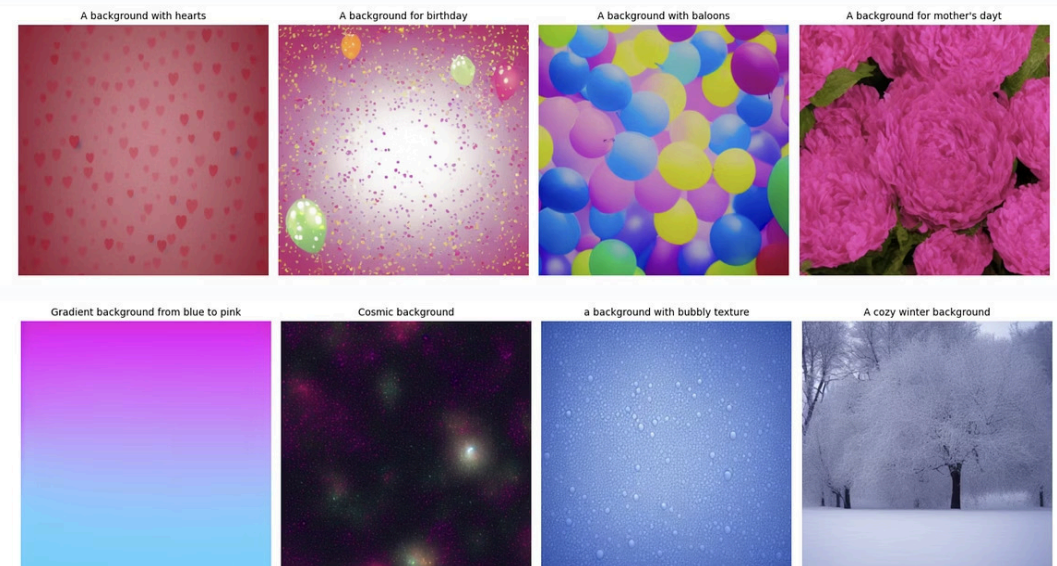
4

Was pushed into Hugging Face Hub

DTD Dataset



Fine-tuned diffuser



Figma Plugin and Layout Design

1 User Interface

Developed a user interface within Figma.

2 Automatic Layout

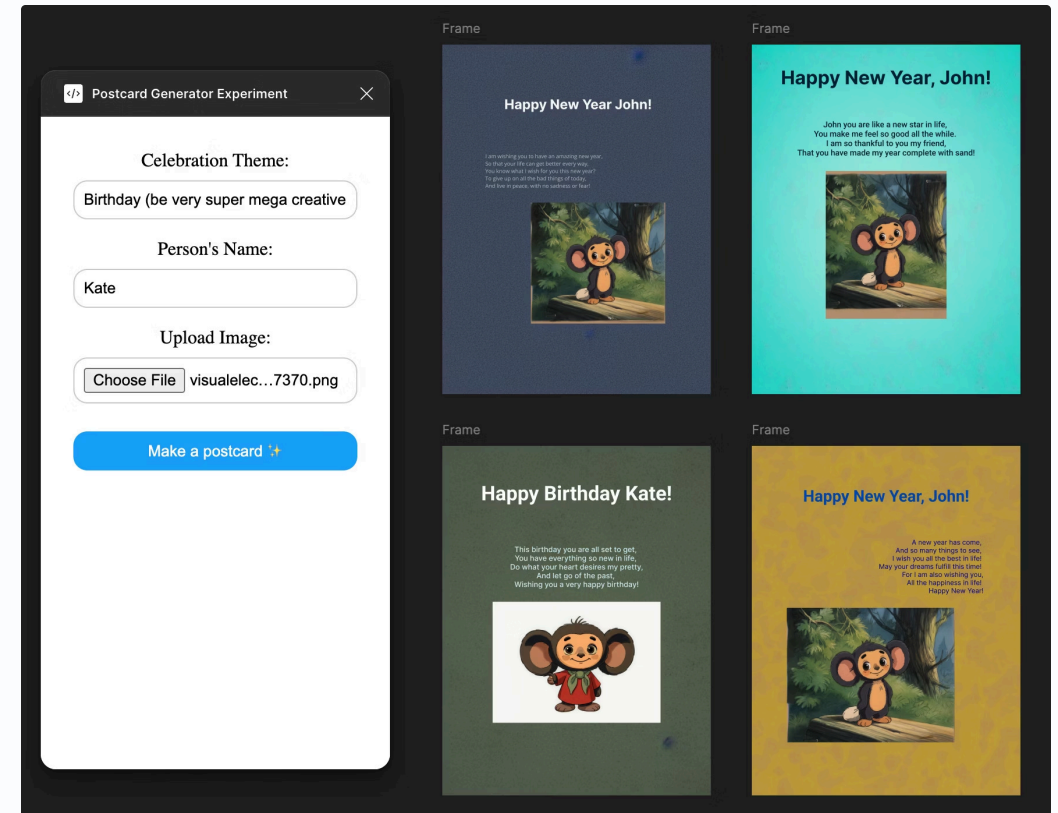
Automatically receive a full greeting card layout within 5 seconds.

3 Customization

Optionally adjust the postcard elements and add new ones.

4 The product was deployed using Modal on free tier

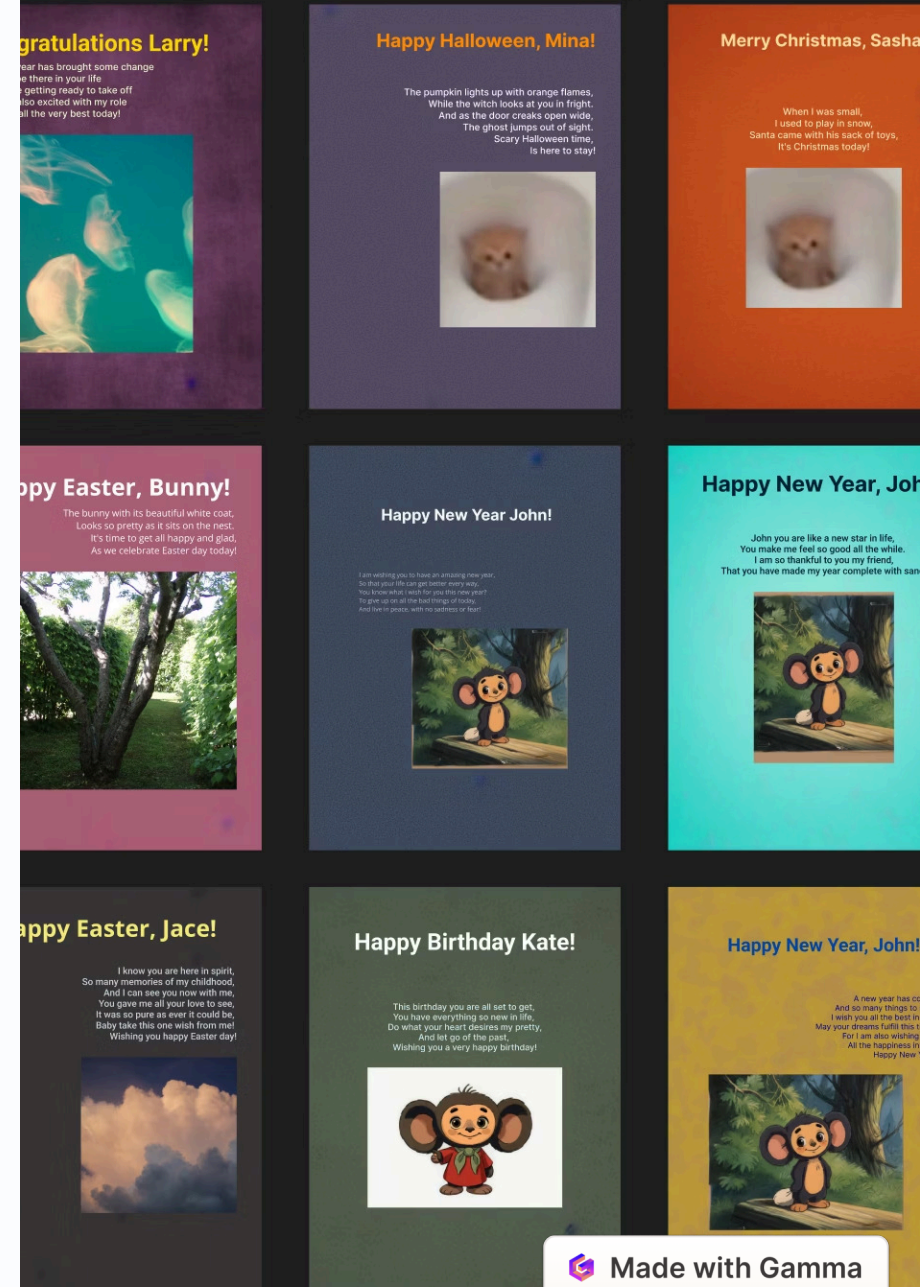
Right now, our plugin is awaiting Figma's official response to be published and used by outer users.



Conclusion

AI GreetCraft, an AI-enhanced Figma plugin, automatically generates personalized greeting cards.

Personalized poems can be efficiently generated using GenAI models fine-tuned on small datasets deployed on Modal. Design integration through the Figma API streamlines user interaction and empowers non-expert creators.



Our Github



Figma plugin link



The link to the more precise report of our work:

<https://github.com/andrewlevada/postcard-generator-experiment/blob/main/Report.pdf>